

UBUNTU SERVER

Enterprise Deployment & Post- Installation Manual

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Target OS:

Ubuntu Server

Platform:

Oracle VM VirtualBox

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Introduction

Operating systems form the core foundation of enterprise IT infrastructure, enabling servers to host web services, databases, and business applications. Before a server can host web applications, databases, cloud services, or internal management systems, it must be properly deployed, configured, secured, and documented. This manual covers the step-by-step guide, for deploying Ubuntu Server 24.04 LTS instance within a virtualized enterprise environment.

Ubuntu Server Installation Guide

Phase 1: Virtual Machine Provisioning

1. Launch Oracle VM VirtualBox and select New to create a virtual machine.
2. Enter the VM Name (e.g., Ubuntu-Server-Week03).
3. Select the target Ubuntu Server ISO Image and ensure unattended installation is disabled (manual installation mode).
4. Allocate 4096 MB (4 GB) of RAM and 2 vCPUs.
5. Create a 40 GB virtual hard disk and attach the virtual network adapter to NAT mode.

❖ **Note:** Account identifiers used in this guide represent baseline standards for this deployment and can be adjusted based on host resources and enterprise naming conventions.

Phase 2: System Installation & Profile Setup

1. Start Ubuntu-Server-Week03 and select Ubuntu Server when prompted for the installation type.
2. Proceed to the Profile Configuration step and input system identity parameters:
 - Your Name: Set administrative name (e.g., YourName)
 - Server Name: Set system hostname (e.g., server01)

- Username: Set account user ID (e.g., quizon)
 - Password: Assign a secure administrative password
 - 3. On the SSH Setup screen, check Install OpenSSH server to enable remote administrative access.
 - 4. Allow the system installer to run until fully completed.
 - 5. Select Reboot Now to restart the virtual machine and unmount the installation media.
- ❖ **Note:** Account identifiers used in this guide represent baseline standards for this deployment and can be adjusted based on host resources and enterprise naming conventions.

Initial Server Configuration

Step 1: System Authentication

1. Log in with the administrative credentials created during setup.

Step 2: System Package Update & Patching

1. Fetch the latest repository updates and apply system patches
2. Type in “sudo apt update && sudo apt upgrade -y”

Server Verification Procedures

1. Verify System Hostname:
 - Type “hostname” to verify that it matches.
2. Verify IP Address Assignment:
 - Type “ip addr” to display the IP.
3. Test Outbound Internet Reachability:
 - Type “ping -c 4 google.com”.
 - Check if there are no packet loss to confirm DNS resolution and internet routing.

Troubleshooting Guide

Issue 1: Large Installation Media Downloads

- **Problem:** Downloading large installation ISOs takes an extended period or stalls due to network drops.
- **Resolution:** Maintain patience and use a stable network connection. Interrupting large transfers causes corrupted ISO files; allowing the download to finish cleanly prevents boot errors during VM creation.

Issue 2: Terminal Repository Mirror Failure (403 Forbidden)

- **Problem:** Executing “sudo apt update” or “sudo apt upgrade -y” returns a 403 Forbidden response while attempting to fetch packages from regional mirrors (ph.archive.ubuntu.com).
- **Resolution:** Resynchronize the package index and run the repair update flag:

Best Practices

1. Make sure you took note of all profile configurations you have made.
2. Keep track of your storage and CPU, running VirtualBox make take a few storage and processing power to run smoothly.
3. Always log in using your standard user account and run administrative commands with “sudo” to prevent accidental system-wide modifications.
4. Avoid closing the VirtualBox window by clicking the X on the top of the window. Always use proper terminal commands like “sudo poweroff” or “sudo shutdown now” to prevent virtual hard disk and file system corruption.

References

VirtualBox Manual: <https://www.virtualbox.org/manual/topics/create-vm.html>